

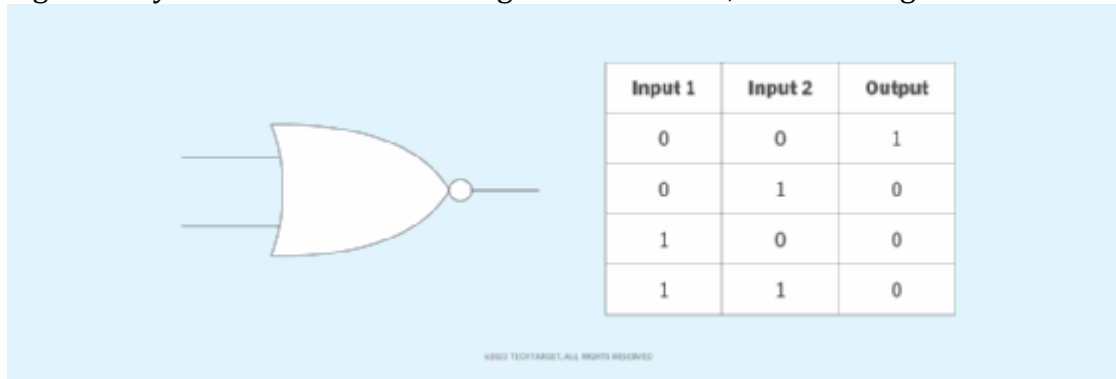
Unit III

Short Question & Answers:

1. Explain classical logic gates NOR, NAND with truth tables.

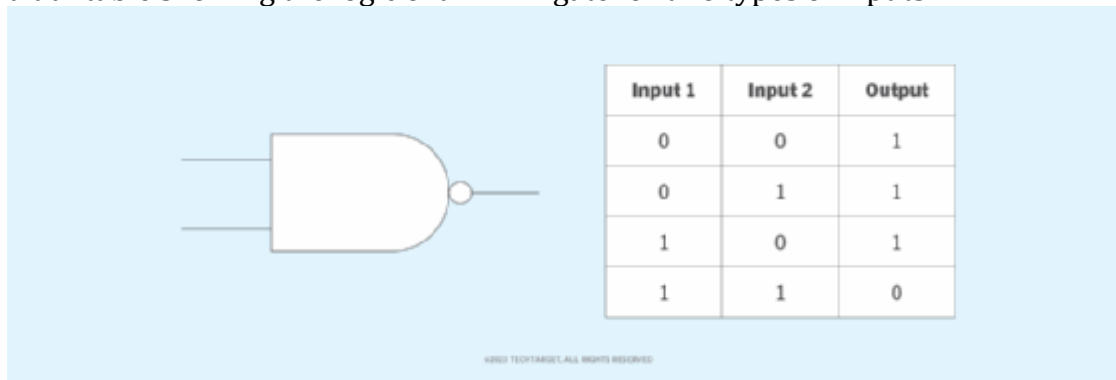
NOR gate

The NOR, or NOT OR, gate is a combination OR gate followed by an inverter. Its output is true or high if both inputs are false or low (logic 0). Otherwise, the output is false. This logic is easy to understand with the gate's truth table, shown in Figure 6.



NAND gate

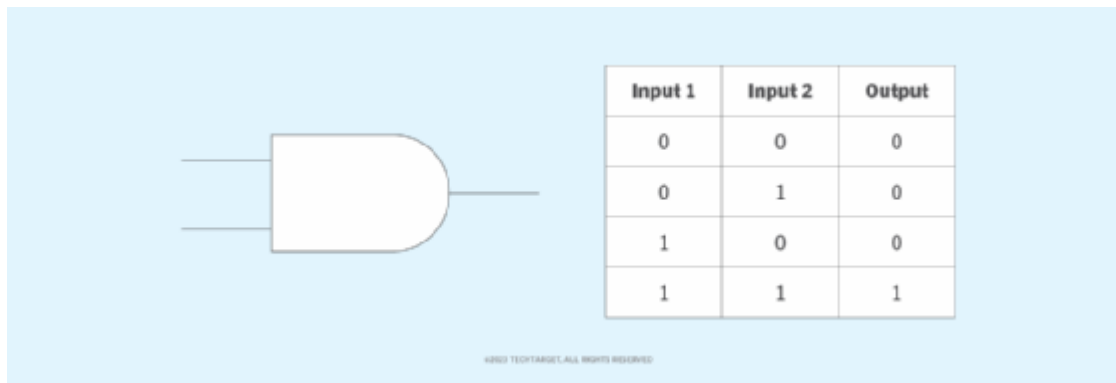
The negated AND, or NAND, gate operates as an AND gate followed by a NOT gate. Its symbol is an AND gate with the circle of a NOT gate at the output. The NAND gate produces a false or low output if both inputs are true. Otherwise, the output is true. Another way to visualize it is that a NAND gate inverts the output of an AND gate. It acts in the manner of the logical operator *and* followed by negation. Figure 5 provides a truth table showing the logic of a NAND gate for two types of inputs.



2. Explain classical logic gates AND, OR with truth tables

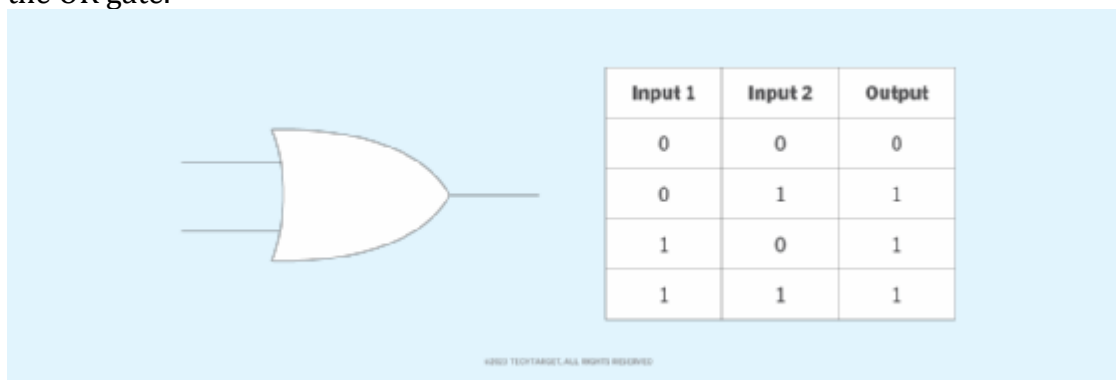
AND gate

The AND gate produces an output of true only when both inputs are true. Otherwise, the output is false. In other words, the output is 1 only when both inputs are 1, and 0 if even one of the inputs is 0. The gate is named so because, if 0 is false and 1 is true, the gate acts in the same way as the logical *and* operator. Figure 1 provides a [truth table](#) showing the circuit symbol and logic combinations for an AND gate. In the symbol, the input terminals are on the left, and the output terminal is on the right.



OR gate

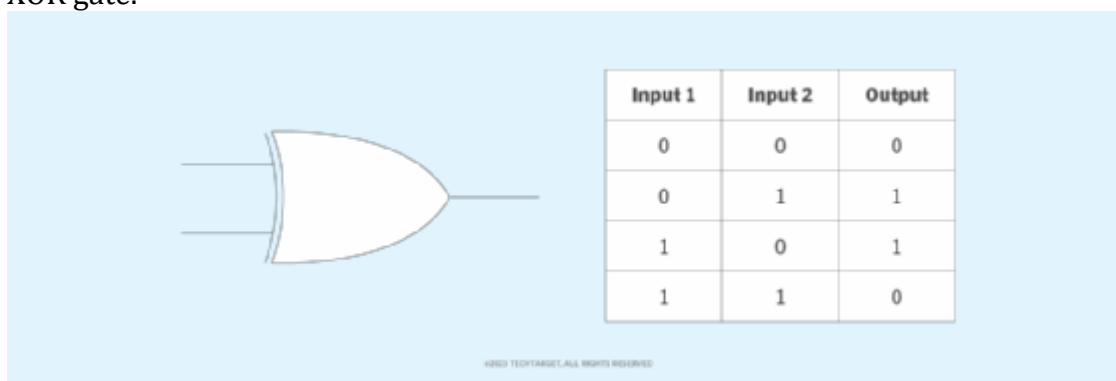
In the OR gate, which is among the most widely used logic circuits in digital devices and [components](#), the output is true if one or both inputs are true. If both inputs are false, the output is false. For the output to be 1, at least one input must be 1. The gate gets its name from behaving like the logical inclusive *or*. Figure 2 shows a truth table for the OR gate.



3.Explain classical logic gates XOR, XNOR with truth tables

XOR gate

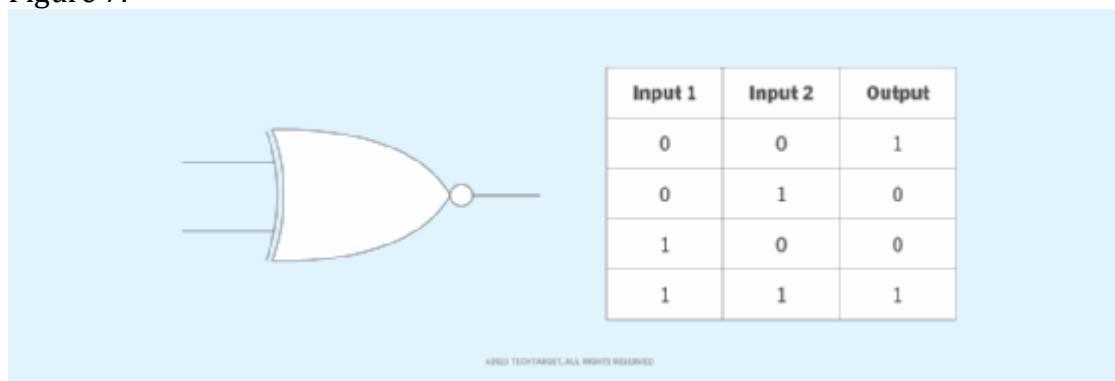
The XOR, or exclusive OR, gate acts in the same way as the logical *either/or*. The output is true if either input is true, but not both. The output is false if both inputs are false or both inputs are true. Similarly, the output is 1 (true or high) if the inputs are different, but 0 (false or low) if the inputs are the same. Figure shows a truth table for a 2-input XOR gate.



XNOR gate

The XNOR, or exclusive NOR, gate is a combination of an XOR gate followed by an

inverter. Its output is true if the inputs are the same and false if the inputs are different. It produces a true output, regardless of whether the inputs are true or false. However, they must both be the same for the output to be true. The XNOR truth table looks like Figure 7.



Long Question & Answers:

4. Write the matrix representations of Hadamard (H) gate

Hadamard Gate (H-gate) – It is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gate and its symbol is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



Action of H gate on inputs:

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|0\rangle$ state

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|1\rangle$ state.

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

If a superposed qubit goes through H gate, the result will be

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}}\right)|0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right)|1\rangle$$

$$H|\psi\rangle = \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Gate representation is

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{H} \longrightarrow \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

The truth table is as follows

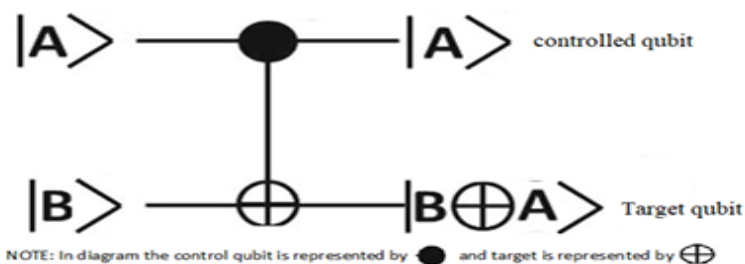
Input	Output
$ 0\rangle$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$
$ 1\rangle$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha\left(\frac{ 0\rangle + 1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{ 0\rangle - 1\rangle}{\sqrt{2}}\right)$

5. Explain the working of the **CNOT (Controlled-NOT)** gate with its circuit and matrix form.

1. Controlled Gate (CNOT)

The CNOT gate is a two-qubit operation, where the first qubit is referred as the control qubit $|A\rangle$ and the second qubit as the target qubit $|B\rangle$. If the control qubit is $|1\rangle$ then it will flip the target qubit state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$. When the control qubit is in state $|0\rangle$ then the target qubit remains unchanged.

The symbolic representation is as follows. The upper line represents control qubit and bottom line represents target qubit.



In the combined qubit, first term is control qubit and the second term is target qubit. For ex, in $|AB\rangle$, A is control qubit and B is target qubit

Matrix form of **CNOT** Gate is given by

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The Transformation could be expressed as $|A, B\rangle \rightarrow |A, B \oplus A\rangle$

Action of the gate: Consider the operations of CNOT gate on the four inputs $|00\rangle, |01\rangle, |10\rangle$ and $|11\rangle$.

- i) Operation of CNOT Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$
Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|0\rangle$.
- ii) Operation of CNOT Gate for input $|01\rangle$: $|01\rangle \rightarrow |01\rangle$
Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|1\rangle$
- iii) Operation of CNOT Gate for input $|10\rangle$: $|10\rangle \rightarrow |11\rangle$
Here, control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|0\rangle$ to $|1\rangle$.
- iv) Operation of CNOT Gate for input $|11\rangle$: $|11\rangle \rightarrow |10\rangle$
Here control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|1\rangle$ to $|0\rangle$.

Input		Output	
$ A\rangle$	$ B\rangle$	$ A\rangle$	$ B \oplus A\rangle$
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$

The Truth Table of operation of CNOT gate is as follows.

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

6. Write the matrix representations of **Pauli-X, Pauli-Y, and Pauli-Z** gates

Single qubit gates

1. X – Gate

This is also called as Pauli X – gate. It is given by

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

When X operates on $|0\rangle$ and $|1\rangle$ the output will be inverted (ie, $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes $|0\rangle$)

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Since X inverts each input it is also called as *bit-flip* gate. If a superposed qubit goes through X gate, the result will be

$$X|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

So,

$$X|\Psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

Symbolically these gates are represented as follows



2. Y – Gate

This is also called as Pauli Y – gate. It is given by

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

When Y operates on $|0\rangle$ and $|1\rangle$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \times 1 + (-i) \times 0 \\ i \times 1 + 0 \times 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \times 0 + (-i) \times 1 \\ i \times 0 + 0 \times 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

So,

$$Y|0\rangle = i|1\rangle \text{ and } Y|1\rangle = -i|0\rangle$$

If a superposed qubit goes through Y gate, the result will be

$$Y|\Psi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \times \alpha + (-i) \times \beta \\ i \times \alpha + 0 \times \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i\beta|0\rangle + i\alpha|1\rangle$$

So,

$$Y|\Psi\rangle = i\alpha|1\rangle - i\beta|0\rangle$$

Symbolically these gates are represented as follows



3. Z – Gate

This is also called as Pauli Z – gate. It is given by

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

When Z operates on $|0\rangle$ and $|1\rangle$ the phase will change. Hence this is also called as *phase-flip* gate

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 0 \times 0 \\ 0 \times 1 + (-1) \times 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 0 \times 1 \\ 0 \times 0 + (-1) \times 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

So,

$$Z|0\rangle = |0\rangle \text{ and } Z|1\rangle = -|1\rangle$$

If a superposed qubit goes through Z gate, the result will be

$$Z|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 \times \alpha + 0 \times \beta \\ 0 \times \alpha + (-1) \times \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

So,

$$Z|\Psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

Symbolically these gates are represented as follows

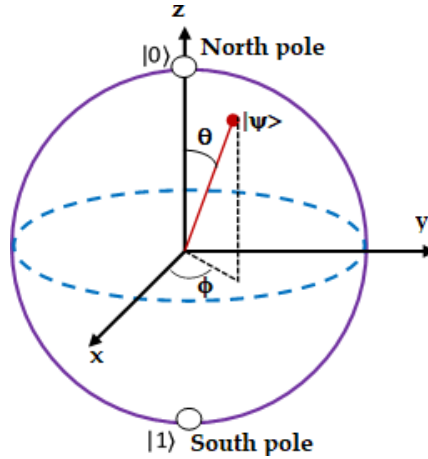


7. Explain the Bloch sphere representation of a qubit with a neat diagram.

Bloch sphere

Bloch sphere is an imaginary sphere which is used to represent pure single-qubit states as a point on its surface. It has unit radius. Its North Pole and South Pole are selected to represent the basis states namely $|0\rangle$ and $|1\rangle$. North Pole represents $|0\rangle$ (say spin up $\uparrow$) and South Pole represents $|1\rangle$ (say spin down $\downarrow$). All other points on the sphere represent

superposed states (ie, state space). Bloch sphere allows the state of a qubit to be represented in spherical coordinates (ie, r , θ and ϕ). It is as follows



The state qubit $|\psi\rangle$ on the Bloch sphere makes an angle θ with z-axis and its projection (azimuth) makes angle ϕ with x-axis as shown. It is clear from the fig that $0 < \theta < \pi$ and $0 < \phi < 2\pi$. $|\psi\rangle$ is represented as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

It can be proved that

$$|\Psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \text{ --- (1)}$$

Using this equation we can represent $|\psi\rangle$ for different θ and ϕ as follows

Case-1: let $\theta = 0$ and $\phi = 0$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos 0 |0\rangle + e^{i0} \sin 0 |1\rangle = |0\rangle + 0 \\ \therefore |\Psi\rangle &= |0\rangle \end{aligned}$$

This is the **north pole**, representing the base state $|0\rangle$.

Case-2: let $\theta = \pi$ and $\phi = 0$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos \frac{\pi}{2} |0\rangle + e^{i0} \sin \frac{\pi}{2} |1\rangle = 0 + |1\rangle \\ \therefore |\Psi\rangle &= |1\rangle \end{aligned}$$

This is the **south pole**, representing the base state $|1\rangle$.

Case-3: let $\theta = \pi/2$ and $\phi = 0$, then eq (1) becomes

$$|\Psi\rangle = \cos \frac{\pi}{4} |0\rangle + e^{i0} \sin \frac{\pi}{4} |1\rangle$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

$$|\Psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

This point lies on the **positive x-axis**, representing the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Case-4: let $\theta = \pi/2$ and $\phi = \pi$, then eq (1) becomes

$$|\Psi\rangle = \cos \frac{\pi}{4} |0\rangle + e^{i\pi} \sin \frac{\pi}{4} |1\rangle$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$$

$$|\Psi\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

This point lies on the **negative x-axis**, representing the superposition $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. In the above discussion we have represented only single qubit state. Bloch sphere is a nice visualization of single qubit states.

The Bloch sphere is a vital visual model that represents a single qubit's state as a point on a sphere, where its position defines the qubit's probability and phase.

Unit IV

Short Question & Answers:

1. What is the main difference between classical circuits and quantum circuits

Classical Circuits	Quantum Circuits
① Basic unit of Classical is Bit (0 or 1)	① Basic unit of Quantum is Qubit (Superposition of 0 and 1)
② State representation of Classical single definite value (0 or 1)	② State representation of Quantum is vector in Hilbert space Eg: $\alpha 0\rangle + \beta 1\rangle$
③ Classical operate based on Boolean algebra and deterministic logic gates AND, OR etc.	③ Quantum Linear algebra matrices, tensors, vectors, Unitary operators.
④ Classical most gates are irreversible	④ Quantum All gates must be reversible.

2. What are the basic notations used in quantum circuits, such as $|0\rangle$, $|1\rangle$, and gate symbols?

Basic circuit notation:

A quantum circuit is a diagrammatic representation of how qubits are processed in a quantum computation.

The basic notations used in circuits so that operations can be easily understood.

⇒ X gate: Flips the qubit

——— $\boxed{X}$ ———

⇒ H gate: Creates superposition.

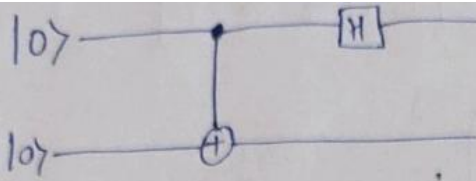
——— $\boxed{H}$ ———

⇒ Z and Y gates: Apply phase and rotation changes.

——— $\boxed{Y}$ ———

——— $\boxed{Z}$ ———

3. Draw the circuit for non-entanglement



A CNOT gate is applied to the $|00\rangle$ state followed by a Hadamard gate on the first qubit.

$\psi_i = |00\rangle$

$\psi_f = (H \otimes I) \text{ CNOT } |00\rangle$

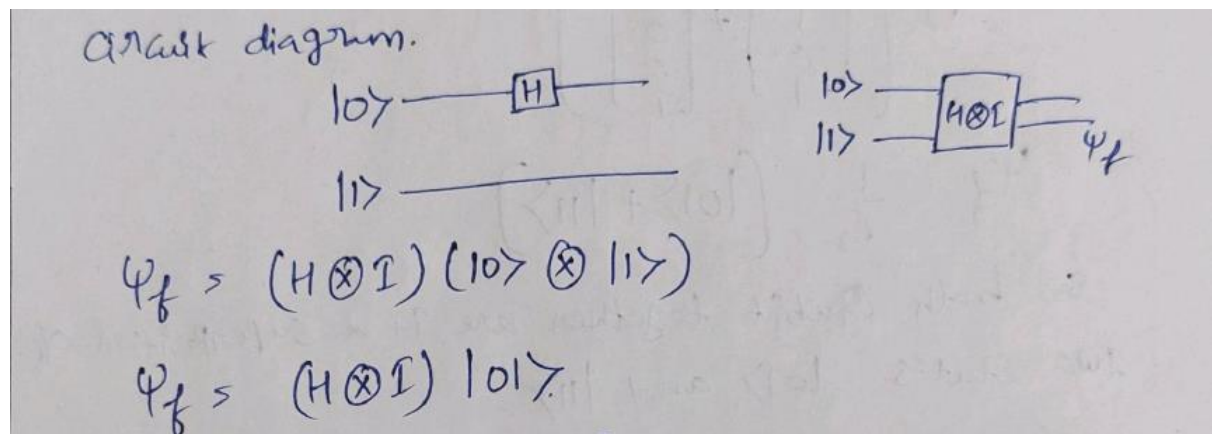
4. Explain the Introduction to quantum circuits

Introduction to Quantum Circuits:

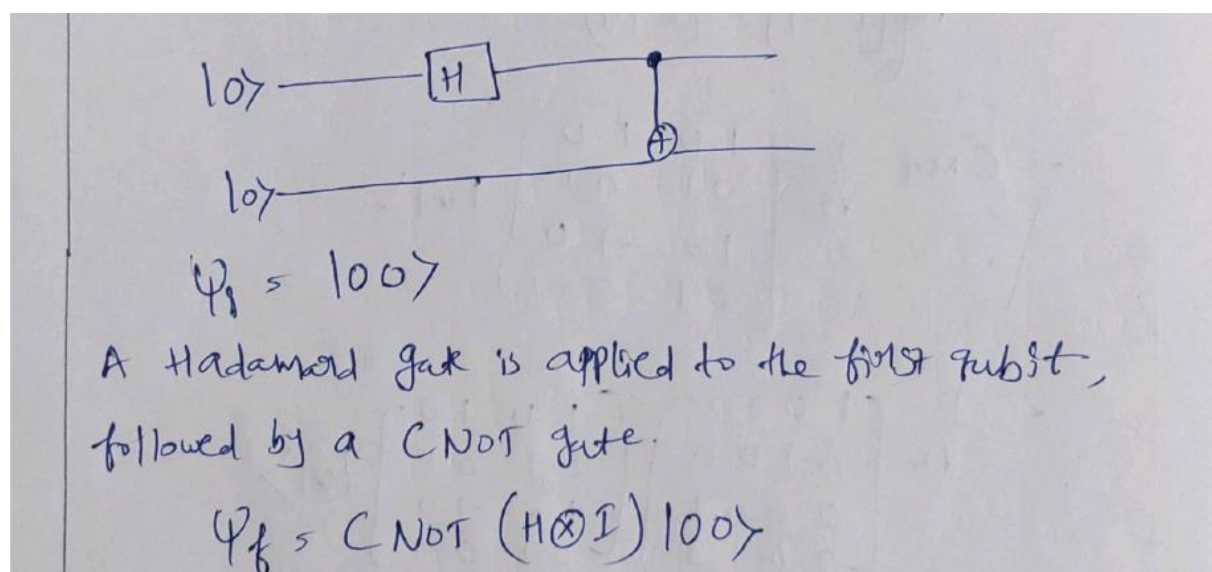
A Quantum Circuit is a sequence of Quantum gates applied to a set of Qubits. Qubits can exist in a superposition of both states simultaneously.

This Property along with Entanglement and Interference, allows Quantum Circuits to perform complex computations more efficiently than classical circuits for certain problems.

5. Draw the circuit diagram to create a superposition using H and NOP



6. Draw the circuit diagram to create a Bell pair using H and CNOT gates.

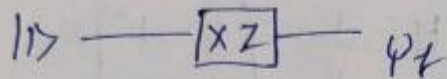
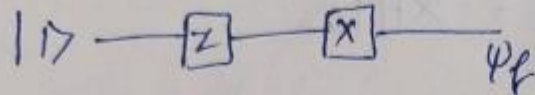


Long Question & Answers:

7. Find the final qubit state for the circuit —[Z]—[X]— Using inputs $|0\rangle$ and $|1\rangle$

Apply Z gate followed by X gate on $|1\rangle$

Consider the circuit



$$\psi_i = |1\rangle, \quad \psi_f = XZ|1\rangle$$

Matrices are

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

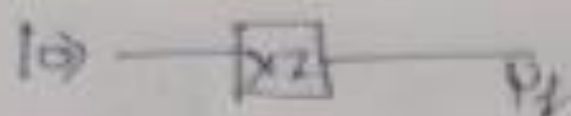
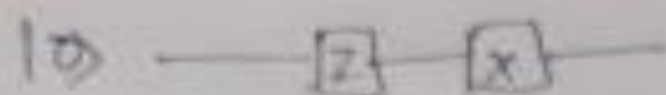
$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$XZ|1\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -|0\rangle$$

$$\psi_f = XZ|1\rangle$$

$$\psi_f = -|0\rangle$$

Z-X on $|0\rangle$



$$\psi_i = |0\rangle \quad \psi_f = XZ|0\rangle$$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

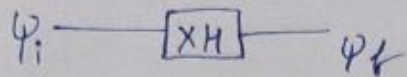
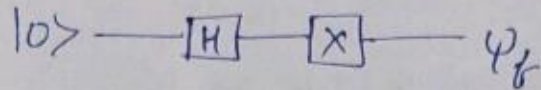
$$XZ|0\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$\psi_f = XZ|0\rangle = |1\rangle$$

8. Find the final qubit state for the circuit —[H]—[X]— Using inputs $|0\rangle$ and $|1\rangle$

Apply Hadamard (H) gate followed by X gate on $|0\rangle$

Circuit diagram.



$$\psi_i = |0\rangle, \quad \psi_f = XH|0\rangle$$

$$\text{Matrices } X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$XH = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$XH|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle + |1\rangle \right]$$

$$\psi_f = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

H Creates a Superposition and X flips the components but keeps the Superposition same.

H-X on $|1\rangle$

$|1\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |X\rangle)$

$\psi_i |1\rangle \Rightarrow \psi_f = XH|1\rangle$

$$XH|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$XH|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

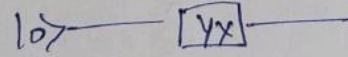
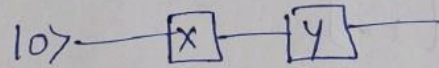
$$= \frac{1}{\sqrt{2}} \left[-\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]$$

$$= \frac{1}{\sqrt{2}} [-|0\rangle + |1\rangle]$$

9. Find the final qubit state for the circuit —[X]—[Y]— Using inputs $|0\rangle$ and $|1\rangle$

Applying X gate followed by Y gate on $|0\rangle$

Consider the circuit



$$\psi_i = |0\rangle, \quad \psi_f = YX|0\rangle$$

matrices of the Pauli gate are

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$YX = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

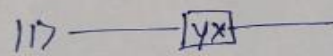
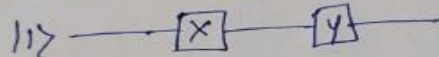
$$YX|0\rangle = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

$$\psi_f = YX|0\rangle$$

$$\psi_f = -i|0\rangle$$

Apply X gate followed by Y gate on $|1\rangle$

Consider the circuit



$$\psi_i = |1\rangle, \quad \psi_f = YX|1\rangle$$

$$\text{matrix } YX = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

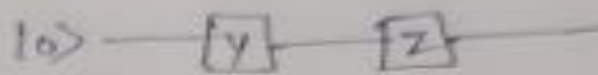
$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

$$YX|1\rangle = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$\psi_f = YX|1\rangle \Rightarrow \psi_f = i|1\rangle$$

10. Find the final qubit state for the circuit —[Y]—[Z]— Using inputs $|0\rangle$ and $|1\rangle$

Apply Y followed by Z on $|0\rangle$



$$\psi_i = |0\rangle$$

$$\psi_f = ZY|0\rangle$$

$$ZY|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} |0\rangle$$

$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} |0\rangle$$

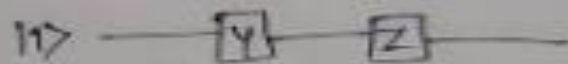
$|1\rangle$

$$\psi_f = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\psi_f = \begin{bmatrix} 0 \\ -i \end{bmatrix} \Rightarrow -i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -i|1\rangle$$

$$\psi_i = |0\rangle \Rightarrow \psi_f = -i|1\rangle$$

Apply 'Y' followed by 'Z' on $|1\rangle$



$$\psi_i = |1\rangle \Rightarrow \psi_f = ZY|1\rangle$$

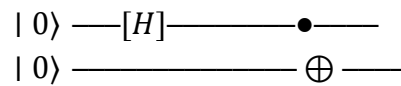
$$ZY|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} |1\rangle$$

$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} |1\rangle$$

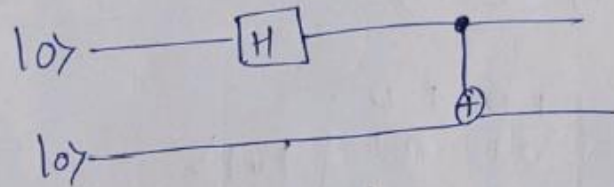
$$\psi_f = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\psi_f = -i|0\rangle$$

11. Find the final state of the following circuit:



Applying H gate and CNOT gate



$$\psi_i = |00\rangle$$

A Hadamard gate is applied to the first qubit, followed by a CNOT gate.

$$\psi_f = \text{CNOT} (H \otimes I) |00\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Psi_f = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} |00\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} |00\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

multiple with $|00\rangle$. This operation
pullout the first column of the matrix

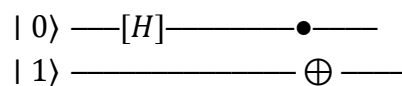
$$\Psi_f = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Psi_f = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right]$$

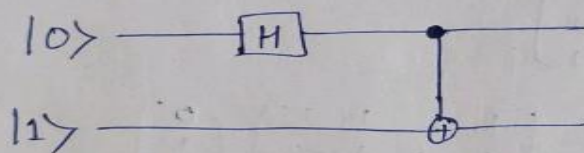
$$\Psi_f = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

This is the Entangled state or Bell state.

12. Find the final state of the following circuit:



Entanglement using Quantum Circuits



The system starts with two separate qubits

→ A Hadamard gate (H) is applied to the first qubit $|0\rangle$.

→ The second qubit $|1\rangle$ is unchanged.

$$\psi_i = |01\rangle$$

$$\psi_f = \text{CNOT}(H \otimes I) |01\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\psi_f = \text{CNOT} \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) |01\rangle$$

$$= \text{CNOT} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$\psi_f = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

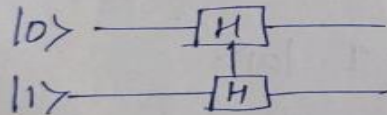
This is a Entangled State known as a Bell State.

13. Find the final state of the following circuit:

$$|0\rangle \text{---} [H] \text{---}$$

$$|1\rangle \text{---} [H] \text{---}$$

Applying a Hadamard (H) gate to Each qubit in a 2-qubit System:



A Hadamard gate is applied to the first qubit, and another H-gate is applied to the second qubit. This parallel operation is represented by the tensor product $(H \otimes H)$.

$$\psi_i = |01\rangle$$

$$\psi_f = (H \otimes H) |01\rangle$$

Using matrix mechanics

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H \otimes H = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \otimes \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & -1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$(H \otimes H) |01\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

The vector $|01\rangle$ only has a '1' in the second position, this operation simply pulls out the second column of the matrix.

$$\begin{aligned} |00\rangle &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ |01\rangle &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \\ |10\rangle &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \\ |11\rangle &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$\psi_f = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\psi_f = \frac{1}{2} \left[\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right]$$

Converting each vector back to its ket representation

$$\psi_f = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

So both qubits together are in superposition of four states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

14. Find the final state of the following circuit:

$|0\rangle \text{ --- } [H] \text{ ---}$

$|1\rangle \text{ ---}$

Superposition using Quantum Circuit :-

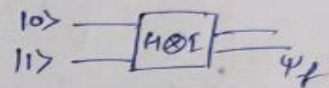
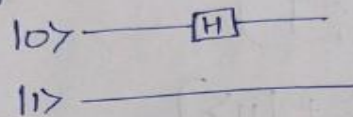
Superposition is a fundamental property of quantum mechanics where a qubit can exist in a combination of basis states $|0\rangle$ and $|1\rangle$ simultaneously.

A Quantum Circuit can be used to create Superposition using the Hadamard gate (H gate).

Consider a 2 qubit system initially in the state $|\psi_i\rangle = |01\rangle$

A Hadamard gate is applied only on the first qubit and second qubit is left unchanged.

Circuit diagram.



$$\psi_f = (H \otimes I) (|0\rangle \otimes |1\rangle)$$

$$\psi_f = (H \otimes I) |01\rangle$$

Matrices $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$H \otimes I |0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes I |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \times I & 1 \times I \\ 1 \times I & -1 \times I \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\psi_f = \frac{1}{\sqrt{2}} (|01\rangle + |11\rangle)$$

So both Qubit together are in a superposition of two states $|01\rangle$ and $|11\rangle$

UNIT V:

Short Question & Answers:

1. Write two applications of quantum computing in cybersecurity

1. Quantum Key Distribution (QKD)

- QKD uses photons to send secret keys securely.
- Any attempt to hack the communication immediately changes the photons, alerting users.

2. Fraud Detection

- Quantum machine learning can detect unusual patterns in financial transactions.
- This helps banks identify fraud faster and more accurately.

2. Mention any two applications of quantum deep learning.

1. Improved Optimization

- Deep learning needs to find the best parameters for accuracy.
- Quantum algorithms help reach these best values faster than classical gradient descent.

2. Big Data Handling

- Quantum parallelism allows quick processing of huge amounts of data.
- This makes training large deep learning models more efficient.

3. Write two ethical applications/concerns in quantum computing.

1. Breaking Current Encryption

- Quantum computers can break existing internet security methods.
- This may make current online data and communications unsafe.

2. Privacy Risks

- Sensitive personal and financial data could be exposed if quantum technology is misused.
- This increases the risk of identity theft and fraud.

4. Mention two applications of quantum artificial intelligence

1. Faster ML Training

- Quantum computers reduce the training time for AI and machine learning models.
- This helps develop AI applications more quickly.

2. Better Search Algorithms

- Grover's algorithm allows faster search through large databases.
- This improves efficiency in AI-based data retrieval.

5. Write two applications of quantum computing in molecular modelling

1. Drug Discovery

- Quantum models study how drug molecules interact with proteins.
- This reduces long lab experiments and speeds up new medicine development.

2. Simulating Chemical Reactions

- Quantum computers can model complex reactions faster than classical computers.
- This helps chemists understand reaction pathways clearly.

6. Describe any two applications of Quantum Neural Networks (QNNs) in real-world problem-solving

1. Medical Diagnosis

- QNNs can analyze medical scans and patient data faster.
- This helps doctors identify diseases more quickly and accurately.

2. Fraud Detection

- QNNs detect unusual or suspicious transactions in real time.
- This helps banks prevent financial fraud effectively.

Long Question & Answers:

7. Explain the applications of quantum computing in artificial intelligence

1. Faster ML Training

- Quantum computers reduce the training time for AI and machine learning models.
- This helps develop AI applications more quickly.

2. Better Search Algorithms

- Grover's algorithm allows faster search through large databases.
- This improves efficiency in AI-based data retrieval.

3. Big Data Analytics

- Quantum AI can process and analyze huge datasets easily.
- This helps in identifying trends and patterns faster.

4. Quantum Robots

- Robots with quantum processors can make smarter and faster decisions.
- This improves their performance in complex environments.

5. Predictive Healthcare

- Quantum AI can detect diseases at an early stage.
- This supports better treatment and preventive care.

6. Optimization (Logistics & Traffic)

- Quantum AI helps in planning routes to reduce travel time and fuel usage.
- It also helps manage traffic congestion efficiently.

7. Climate Modelling

- Quantum AI predicts weather and climate patterns with high accuracy.
- This helps in disaster preparedness and environmental planning.

8. Real-Time Language Translation

- Quantum AI enables faster and more accurate language translation.
- This improves communication in global applications.

9. Autonomous Drones

- Quantum AI helps drones navigate safely and avoid obstacles.
- This improves efficiency in delivery and surveillance tasks.

10. Smart City Systems

- Quantum AI helps manage electricity, traffic, and surveillance in cities.
- This leads to better control and more efficient urban living.

8. Explain the major applications of quantum computing in molecular modelling and new material design.

Molecular Modelling Applications:

1. Predicting Molecular Structures

- Quantum computers simulate atoms and electrons with high accuracy.
- This helps scientists understand the exact shape and behavior of molecules.

2. Drug Discovery

- Quantum models study how drug molecules interact with proteins.
- This reduces long lab experiments and speeds up new medicine development.

3. Simulating Chemical Reactions

- Quantum computers can model complex reactions faster than classical computers.
- This helps chemists understand reaction pathways clearly.

4. Protein Folding Studies

- Quantum systems can predict protein shapes accurately.
- This is useful for understanding diseases caused by misfolded proteins.

New Material Design Applications:

5. Designing Better Batteries

- Quantum computing helps find materials that store more energy.
- It supports the development of long-lasting and fast-charging batteries.

6. Superconductor Design

- Quantum computers predict materials that conduct electricity with zero resistance.
- This helps in creating efficient power and electronic systems.

7. New Polymer Materials

- Quantum models help design strong, lightweight polymer materials.
- These materials are useful in aerospace and engineering industries.

8. Catalyst Optimization

- Quantum simulations help industries find catalysts that speed up reactions.
- This reduces energy use and improves production efficiency.

9. Predicting Material Properties

- Quantum computers can predict properties like magnetism and conductivity.
- This helps scientists design materials for specific uses.

10. Carbon Capture Materials

- Quantum models help find materials that absorb CO₂ more effectively.
- This supports climate control and pollution-reduction efforts.

9. Explain the role of quantum computing in improving cybersecurity applications

1. Post-Quantum Cryptography

- New encryption methods are being developed to resist quantum computer attacks.
- These methods keep data safe even when powerful quantum systems appear.

2. Breaking Classical Encryption

- Quantum algorithms like Shor's can break RSA and ECC very quickly.
- This shows that current classical encryption is no longer fully safe.

3. Quantum Key Distribution (QKD)

- QKD uses photons to send secret keys securely.
- Any attempt to hack the communication immediately changes the photons, alerting users.

4. Fraud Detection

- Quantum machine learning can detect unusual patterns in financial transactions.
- This helps banks identify fraud faster and more accurately.

5. Protecting Critical Systems

- Quantum security protects government, military, and power grid networks.
- This keeps important national systems safe from cyber-attacks.

6. Quantum-Secure Cloud

- Data stored in the cloud becomes extremely hard to hack with quantum encryption.
- This ensures high-level protection for sensitive information.

7. Blockchain Security

- Quantum hashing strengthens blockchain networks.
- This makes digital currencies and records more secure.

8. Intrusion Detection

- Quantum systems can analyze network traffic in real time.
- This allows quick detection of cyber-attacks and unusual activities.

9. Secure Transactions

- Quantum security ensures safety during bank transfers and online payments.
- This reduces the chances of hacking and financial theft.

10. Authentication Systems

- Quantum randomness helps create passwords that are impossible to guess.
- This improves login security for devices and online accounts.

10. Discuss the applications of quantum computing in deep learning

1. Faster Training of Deep Models

- Quantum computers can process many possibilities at the same time.
- This helps deep learning models train much faster than on normal computers.

2. Improved Optimization

- Deep learning needs to find the best parameters for accuracy.
- Quantum algorithms help reach these best values faster than classical gradient descent.

3. Big Data Handling

- Quantum parallelism allows quick processing of huge amounts of data.
- This makes training large deep learning models more efficient.

4. Better Image Classification

- Quantum algorithms recognize patterns more clearly.
- This improves the accuracy of identifying images and objects.

5. Enhanced NLP Models

- Natural Language Processing becomes faster and more accurate with quantum computing.
- It helps in better translation, summarization, and text understanding.

6. Quantum Reinforcement Learning

- Quantum systems help agents make faster decisions.
- This is useful for robots and AI systems that learn from their environment.

7. Recommendation Systems

- Quantum models quickly analyze user preferences.
- This improves suggestions on platforms like YouTube, Netflix, or Amazon.

8. Medical Imaging

- Deep quantum models can detect tumors and diseases faster.
- This helps doctors diagnose illnesses more accurately.

9. Financial Forecasting

- Quantum computing finds complex patterns in market data.
- This helps make better predictions for stocks and investments.

10. Robotics

- Quantum deep learning helps robots understand their surroundings in real time.
- This improves their ability to learn tasks and adapt quickly.

11. Describe the applications of Quantum Neural Networks (QNNs) in real-world problem-solving.

1. Pattern Recognition

- QNNs can detect hidden patterns in data quickly.
- This is useful for analyzing images and identifying trends.

2. Image & Speech Recognition

- Quantum states allow QNNs to process large data efficiently.
- This improves the speed and accuracy of recognizing images and speech.

3. Medical Diagnosis

- QNNs can analyze medical scans and patient data faster.
- This helps doctors identify diseases more quickly and accurately.

4. Fraud Detection

- QNNs detect unusual or suspicious transactions in real time.
- This helps banks prevent financial fraud effectively.

5. Weather Prediction

- QNNs can handle large amounts of climate data.
- This helps predict weather patterns and extreme events more accurately.

6. Autonomous Driving

- QNNs help self-driving cars make decisions quickly.
- This improves safety and efficiency on the roads.

7. Supply Chain Optimization

- QNNs predict delivery routes and plan logistics efficiently.
- This reduces delays and improves overall supply chain management.

8. Language Understanding

- QNNs improve translation, chatbots, and virtual assistants.
- This helps machines understand and respond to human language better.

9. Robotics Control

- QNNs enhance robot learning and movement control.
- This allows robots to perform tasks more accurately and safely.

10. High-Speed Classification

- QNNs can sort and categorize large data sets extremely quickly.
- This is useful for data analysis in research, business, and AI applications.

12. Write ethical applications/concerns in quantum computing.

1. Breaking Current Encryption

- Quantum computers can break existing internet security methods.
- This may make current online data and communications unsafe.

2. Privacy Risks

- Sensitive personal and financial data could be exposed if quantum technology is misused.
- This increases the risk of identity theft and fraud.

3. Power Imbalance

- Only wealthy countries or companies may control quantum computing.
- This could create inequality in technology access and global power.

4. Job Loss

- Automation using quantum computing may reduce classical computing jobs.
- Some traditional IT and data-processing roles may become obsolete.

5. Unregulated AI Behavior

- Quantum AI systems could make biased or harmful decisions.
- Without proper rules, this can affect society negatively.

6. Military Misuse

- Quantum computing could be used for advanced warfare or spying.
- This raises serious security and ethical concerns globally.

7. Data Misuse

- Sensitive information may be abused without proper laws.
- This includes personal, medical, or financial data exploitation.

8. Bias in Quantum Machine Learning

- Quantum ML algorithms can inherit biases from their training data.
- This may lead to unfair decisions in applications like hiring or loans.

9. Health and Genetic Research Risks

- Quantum simulations could be misused for unethical medical or genetic experiments.
- This raises moral and legal concerns about human rights.

10. Environmental Impact

- Quantum computers require cooling systems that consume a lot of energy.
- This can negatively affect the environment if not managed sustainably.